Environmental Product Declaration

In accordance with ISO 14025:2006 for:

[*Continuous Casting Billet*]

from

[*Jiangsu Longteng Special Steel Group Co., Ltd.*]

龙腾特钢集团
LONGTENG SPECIAL STEEL GROUP

Programme:	The International EPD System, www.environdec.com
Programme operator:	EPD International AB
Type of EPD:	EPD of a single product from a manufacturer provider
EPD registration number:	EPD-IES-0025968
Version date:	2025-12-29
Validity date:	2030-12-29

An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com

GENERAL INFORMATION

Programme Information	
Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
E-mail:	support@environdec.com

Product Category Rules (PCR)
Product Category Rules (PCR): ***<2015:03: Basic iron or steel products & special steels, except construction products, Version 3.0.0 and UN CPC code 41121>***
PCR review was conducted by: ***< The Technical Committee of the International EPD® System. See www.environdec.com/TC for a list of members. Review chair: Claudia A. Peña, University of Concepción, Chile. The review panel may be contacted via the Secretariat www.environdec.com/contact >***

Third-party Verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ **Individual EPD verification without a pre-verified LCA/EPD tool** Third-party verifier: *Bureau Veritas Certification Sverige AB* Accredited by: *Bureau Veritas Certification Sweden AB, accredited by SWEDEC with accreditation number 1236* Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.

For further information about comparability, see ISO 14025.

INFORMATION ABOUT EPD OWNER

Owner of the EPD:

Jiangsu Longteng Special Steel Group Co., Ltd.

Address:

No. 118, Hualian Road, Tonggang Industrial Park, Meili Town, Changshu City, Jiangsu Province, China

Contact:

Xiaodan Qiu, qxd@lttg.cn

Description of the organisation:

Jiangsu Longteng Special Steel Group Co., Ltd. (hereinafter referred to as "Longteng") was established in 1994. In 2007, Longteng began to transform and upgrade, by extending the industrial chain and improving the product quality, it gradually becomes a steel intergrated enterprise that covers sintering, ironmaking, steelmaking, refining, continuous casting, rolling, and deep processing of metal products. Longteng sticks to the philosophy of “green special steel and ecological Longteng”, adheres to differentiated product positioning and deep cultivation of steel subdivision industry for years, coupling development of "specialized delicate special innovative" products.

Longteng is a modern specialty steel enterprise that has deeply embedded green, low-carbon development and circular economy principles into its strategy and daily operations. The company has established and operates an integrated green production system covering water, gas, energy, and materials. Key achievements include implementing a closed-loop water management system that achieves zero discharge of production wastewater by fully treating and reusing 100% of industrial water, and advancing high-efficiency energy conversion through initiatives such as independently developing and commissioning China’s first 60MW subcritical gas-fired power unit to convert steelmaking by-product gases into clean electricity, constructing an on-site rooftop photovoltaic power generation system operating under a “self-consumption with surplus grid feed-in” model, and utilizing a 115-ton ultra-high-power electric arc furnace—a short-process core production asset primarily fueled by scrap steel—which reduces energy consumption per ton of steel by over 60% and carbon emissions by more than 70% compared to the traditional integrated route.

Longteng has successively passed multiple system certification, include ISO9001, ISO14001, ISO45001, ISO50001 and so on. The products of Shipbuilding sections, Ingots and Billets have been confirmed by the nine major global classification societies in China, the United States, the United Kingdom, France, Japan, South Korea, Italy, Norway, etc. Among these products, the Ingot has passed the PED product and factory certification of German TUV.

Longteng actively promotes green development, adheres to the green development concept of recycling, energy saving and carbon reduction, and is committed to building a green special steel enterprise that covers the entire process from procurement to final products. Longteng aims to become an advanced enterprise in the industry, by actively benchmarking and learning from excellent peers, updating equipment and technology, increasing investment in energy conservation and environmental protection, and clean production, comprehensively promotes its green development.

PRODUCT INFORMATION

Product name:

Continuous casting billet

Product identification:

GB/T 15574-2016 Steel products classification, GB/T 700-2006, GB/T 711-2017, GB/T 591-2018, GB/T 713.2-2023, GB/T 3077-2015

UN CPC code:

41121

Product description:

The continuous casting billets is primarily produced at Changshu Longteng Special Steel Co.,Ltd. The main sales channels include downstream rolling processes within the company and external sales to other rolling or forging enterprises.

The continuous cast products mainly include square and round cross-sections. The specific dimensional specifications are as follows:

Type	Dimensional specifications
Square Billets	120 x 120 mm 150 x 150 mm 180 x 180 mm 220 x 220 mm 250 x 250 mm
Rectangular Blooms	160 x 230 mm 165 x 280 mm 170 x 280 mm 200 x 310 mm 180 x 220 mm 310 x 360 mm 200 x 380 mm 230 x 350 mm
Round Billets	Φ300mm

The continuous cast billets encompass the following quality grades: structural steel, carbon steel, alloy steel for quenching and tempering, low-alloy steel, case-hardening steel, boron steel, pressure vessel steel, and special steel. Their end-use applications are extensive, spanning downstream sectors including hot-rolled sections, machinery and equipment manufacturing, and the automotive industry.

Figure1 Examples of Continuous cast billets produced by Longteng

Brief Introduction to the Manufacturing Process:

1.Electric Arc Furnace (EAF/EBT) Melting

Raw Material: Scrap steel.

Process: Qualified molten steel is produced through operations such as electrical melting, oxygen blowing, and slag formation, with preliminary alloying during tapping.

2.Ladle Furnace (LF) Refining

Process: The tapped steel is further treated in the LF furnace for deep deoxidation, desulfurization, temperature raising, and precise fine-tuning of alloy composition.

3.Vacuum Degassing (VD)

Process: The refined steel from the LF furnace is transferred to the VD furnace for vacuum treatment to remove harmful gaseous elements such as hydrogen, nitrogen, and oxygen.

4.Continuous Casting

Process: The qualified molten steel is cast, solidified in a continuous caster, and cut into billets/blooms of specified shapes and lengths.

5.Finishing and Warehousing

Process: The cast products undergo necessary finishing, are inspected for quality, weighed, and placed into storage.

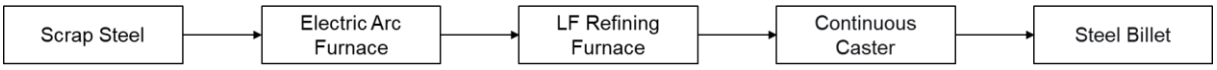


Figure2 Production Process Flow Diagram

Name and location of production site(s):

No. 118, Hualian Road, Tonggang Industrial Park, Meili Town, Changshu City, Jiangsu Province, China.

Website:

For more information or explanatory materials, reference to https://www.lttg.cn/

CONTENT DECLARATION

Product content	Mass, kg	Post-consumer recycled material, mass-% of product	Biogenic material, mass-% of product	Biogenic material, kg C/product or declared unit
Fe	9.53 E+02	Unknown*	0.00	0.00
C	3.91	0.00	0.00	0.00
Si	4.79	0.00	0.00	0.00
Mn	6.02	0.00	0.00	0.00
P	3.22	0.00	0.00	0.00
S	3.19	0.00	0.00	0.00
Cr	3.56	0.00	0.00	0.00
others	22.4	0.00	0.00	0.00
Total	1.00E+03	0.00	0.00	0.00

* In accordance with the GPI provision stating: "If the share of biogenic/recycled material is unknown, this part of the content declaration can be left out or be declared as 0% (a conservative estimate) or unknown," the post-consumer scrap content in the raw materials is indicated as "unknown" in the table above. Detailed allocation procedures are described in the LCA information → Allocation Procedures → Scrap Allocation section.

Packaging materials	Mass, kg	Mass-% (versus the product)	Biogenic material, kg C/product or declared unit
Uninvolved	0.00	0.00	0.00

At the time of data collection, the products do not contain any of the substances included in the Candidate List of Substances of Very High Concern (SVHC) for authorization under the Regulation (EC) No 1907/2006 (REACH), neither above the threshold for registration with the European Chemicals Agency or above 0.1% (wt/wt).

LCA INFORMATION

Declared unit:

1 tonne (1000 kg) of continuous casting billet produced by EAF process.

Reference service life and its relationship with the technical/actual lifespan:

Not applicable

Time representativeness:

The LCA is based on production data from 2024 calendar year.

Geographical scope:

Module A1 and A3 Material suppliers are Global.

Module A3 production is located in China.

Database(s) and LCA software used:

The LCA software is SimaPro V9.6.0.1 and the database is Ecoinvent 3.10.

Description of system boundaries:

This is a Cradle to Gate with modules A1-A3.

Process flow diagram:

Continuous casting billet is produced exclusively via the EAF route, utilizing scrap steel as the raw material, with approximately 2% alloying elements added to enhance mechanical properties. The production is based on an ultra-high-power EAF and integrated with supporting refining and continuous casting equipment. The specific workflow is as follows:

1.EAF Smelting

An ultra-high-power EAF equipped with the Danieli Scrap Preheating Continuous Feeding System (ECS) is used, enabling horizontal continuous feeding suitable for scrap steel of varying densities.

The process integrates full-process foaming slag technology, water-cooled wall and roof slag coating protection, eccentric bottom tapping (EBT) for slag-free tapping, and hot heel practice, effectively improving thermal efficiency and extending furnace lining life.

After power-on, fluxes and alloys are added according to the process requirements. Qualified molten steel is produced through operations such as oxygen blowing and slag formation, with preliminary alloying completed simultaneously during tapping.

2.LF Refining

The molten steel is transferred to the LF furnace for deep refining, including deoxidation, desulfurization, temperature raising, and fine-tuning of alloy composition.

The LF furnace is equipped with a water-cooled roof, a pneumatic sample delivery system, an online chemical composition transmission system, and electrode anti-oxidation coating technology.

By optimizing the Ca-Al slag system and deoxidation practice, the slag-steel interface reaction is intensified to achieve ultra-low sulfur control (sulfur content in steel < 30 ppm).

3.Vacuum Degassing (VD)

After LF refining, the molten steel is treated in the VD furnace. A double-lid, double-tank structure is employed, with the tank lid driven smoothly by a variable-frequency motor.

The system is equipped with vacuum exhaust gas temperature detection instruments, and the vacuum pipeline features a two-stage dust removal system to prevent clogging.

The working vacuum is < 67 Pa, with online hydrogen detection < 2.0 ppm, effectively removing gaseous elements such as H, N, and O.

4.Continuous Casting

Qualified molten steel is cast using full-protection pouring through an 8-strand billet continuous casting machine with a 12-meter radius.

Product specifications include square billets (e.g., 150×150 mm, 180×150 mm) and round billets (e.g., φ300 mm).

The process integrates a large-capacity tundish, mold electromagnetic stirring (M-EMS) + final electromagnetic stirring (F-EMS), automatic mold level control, air-mist secondary cooling, and an intelligent automatic slab quality assessment system, ensuring high purity and excellent microstructural homogeneity.

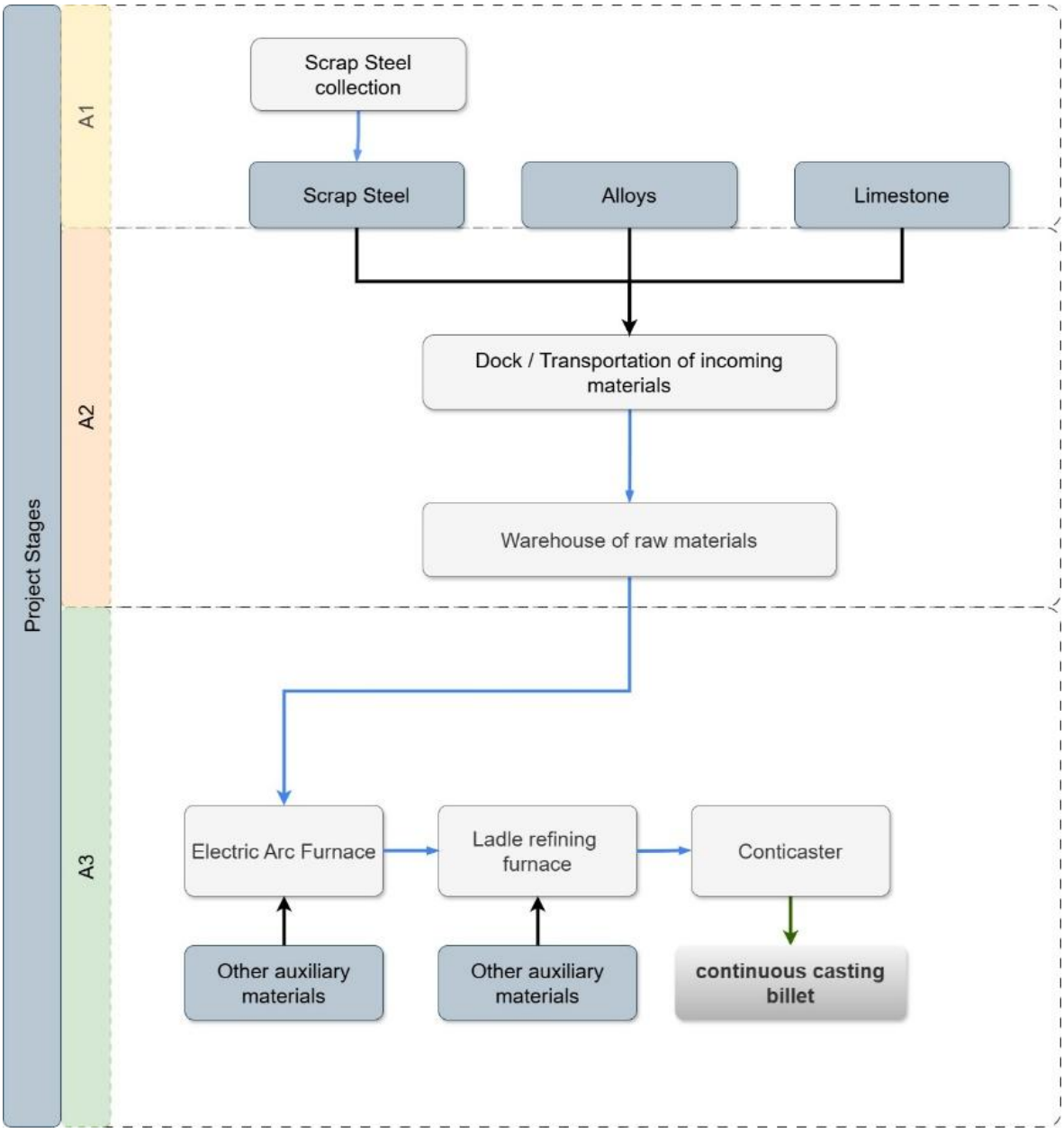
The cast strands are cut to length to form the final continuous cast billets.

5.Finishing and Warehousing

The billets undergo surface conditioning, dimensional inspection, non-destructive testing, and other finishing processes before being marked, weighed, and stored.

This process comprehensively covers the entire production chain from scrap preheating, EAF smelting, secondary refining, vacuum treatment, to continuous casting. It is suitable for producing various high-grade steel types such as structural steel, carbon steel, and alloy steel. The final products are intended for high-end applications including hot-rolled sections, machinery manufacturing, and the automotive industry.

System diagram:



The system boundary in LCA defines the unit processes included in the modeling system. A unit process refers to one or more operations within a manufacturing system. These processes are categorized into modules according to the General Programme Instructions (GPI) for the International EPD® System Version 5.0.1 and the relevant PCR. This analysis follows a cradle-to-gate scope (stages A1–A3). According to the PCR guidance document, the stage categories are defined as follows:

A1 - "Raw Material Extraction and Processing": This category includes the upstream production of steel raw materials as well as the manufacturing of alloy materials. (In this study, the acquisition process of all raw materials entering the final product and co-producr is classified under stage A1, while the acquisition of other auxiliary materials is classified under stage A3.) In accordance with GPI A.3.1 requirements - “Also, each module shall include the upstream production and transport of such waste, i.e., any environmental burden related to a loss shall be included in the module in which the loss occurs.”

- the environmental impacts associated with raw material losses during the production process are consistently assigned to Module A3.

A2 - "Transport of Raw Materials to Manufacturing Facility": This category covers the transportation of all externally sourced raw and auxiliary materials to the plant. In accordance with GPI requirements, both internal material transfers and outbound waste transportation are integral to the production process. Consequently, their associated environmental impacts are allocated to Module A3.

A3 - "Manufacturing Process": This category includes upstream extraction and transportation of lost raw materials, production and use of auxiliary materials (e.g., refractories, lubricants, electrodes), procurement and consumption of energy (e.g., electricity, natural gas, mobile fuels), water consumption, waste treatment and transport, process emissions (including both organized and fugitive gas emissions).

More information:

1 Allocation Procedures

Allocation is made according to the proportion of the target product's output to the total output of the entire factory. During the calculation process, the consumption of energy/resources and the generation of waste, etc., are all apportioned based on this principle.

(1) Facility-Level Allocation

It’s irrelevant, as all production occurs within the same facility.

(2) Co-Product Allocation

Processes may generate by-products or intermediate products beyond the reference flow. These leave the system and are used elsewhere, carrying part of the environmental burden. To allocate burdens meaningfully, the following methods were applied:

During the EAF steelmaking process, continuous casting billets are produced alongside co-products such as EAF slag and waste heat steam. The EAF slag is sold to external plants as a co-product for use in other manufacturing processes, representing a beneficial output. The waste heat is also exported as a co-product to other workshop processes for utilization.

According to the PCR requirements, allocation of inputs and outputs among continuous casting billets, EAF slag, and waste heat steam is performed based on energy demand and other physical relationships. The allocation of exported waste heat steam is based on its calorific value as a proportion of the total heat input. Meanwhile, allocation between EAF slag and continuous casting billets is based on their economic value, as the significant difference in economic value between the two makes physical mass allocation impractical.

(3) Scrap Allocation

The primary raw material for continuous casting billet is purchased scrap. Following the "polluter pays principle", upstream environmental impacts of scrap are not included. Only inbound transport of scrap and any on-site energy or fuel consumption for sorting and handling were considered. As the requirement of the PCR: ’Recycled materials from a scrapyard where the origin is unknown (e.g, data/statistics are missing for the specific scrapyard or the country of its location), shall be assumed to be waste and allocated accordingly.’ In the current calculation model, the scrap sourced from scrapyard is assigned no environmental burden from the previous product system in which the steel scrap was originally produced.

(4) End-of-Life Allocation

No end-of-life allocation was applied, as the system boundary is cradle-to-gate. Background data were truncated without extending to use or disposal phases.

2 Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

Module	Product stage		
	Raw material	Transport	Manufacturing
	A1	A2	A3
Modules declared	X	X	X
Geography	GLO	GLO	CN
Specific data used	74.30%		
Variation-products	0%		
Variation-sites	0%		

3 Electricity mix

The grid electricity mix for the Longteng facility is based on the consumption mix of the State Grid East China Branch (ECGC). This has been modelled using the ecoinvent dataset “Electricity, high voltage {CN-ECGC} | market for electricity, high voltage | Cut-off, U.” The data reflects the 2021-2024 period, as available in the ecoinvent database.

In accordance with the PCR, this electricity profile is classified as Level 3: residual electricity mix on the market - defined as the market consumption mix minus the renewable electricity portion of that mix.

Based on the ecoinvent dataset, the ECGC mix is dominated by hard coal, followed by natural gas and oil. The resulting GHG-GWP indicator for the applied CN-ECGC residual electricity mix is 0.7870 kg CO_2e per kWh.

4. Cut-off criteria

No cut-off criteria were redefined for this study. All known energy and material flow data falling within the system boundary have been included. All relevant life cycle impacts have been considered in the model.

5.Data quality

Data quality requirements are based on the ISO 14040:2006 and ISO 14044:2006 standards. Specific data has been taken on the amounts of materials and energy used during the life cycle of the product. These data are collected, calculated and provided by Longteng, referring to the calendar year 2024. The quality of the specific data is consistent with the standards used. The data was examined carefully, and clarifications requested from the Manufacturer when necessary.

Generic data have been taken on the impact per unit of matter or energy. These data have been obtained from the Ecoinvent database version 3.11.

Process	Source type	Source	Reference year	Data category	Share of primary data,of GWP-GHG results for A1-A3
Raw material extraction and processing	Collected data; Database	EPD owner, Ecoinvent V3.11	2024	Secondary data	0.00%
Transport of all materials from the suppliers to the manufacturer	Collected data; Database	EPD owner, Ecoinvent V3.11	2024	Primary data	0.70%
Manufacturing of the product	Collected data; Database	EPD owner, Ecoinvent V3.11, Electricity modelled by the sub-national consumption mix of the market based on Ecoinvent V3.11	2021-2024	Primary data	73.60%
Total share of primary data,of GWP-GHG results for A1-A3					74.30%

The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

ENVIRONMENTAL PERFORMANCE

Impact category indicators

PARAMETER		UNIT	A1	A2	A3	TOTAL
Global warming potential (GWP)	Fossil	kg CO_2 eq.	1.78E+02	4.84E+00	5.11E+02	6.94E+02
	Biogenic	kg CO_2 eq.	2.88E-01	1.24E-03	1.97E-01	4.86E-01
	Land use and land transformation	kg CO_2 eq.	2.51E-01	2.42E-03	2.77E-01	5.30E-01
	TOTAL	kg CO_2 eq.	1.79E+02	4.84E+00	5.11E+02	6.95E+02
Ozone layer depletion (ODP)		kg CFC 11 eq.	1.30E-06	6.92E-08	1.26E-06	2.63E-06
Acidification potential (AP)		mol H^+ eq.	1.26E+00	1.68E-02	2.12E+00	3.40E+00
Eutrophication potential (EP)	Aquatic freshwater	kg P eq.	5.51E-02	5.62E-04	8.52E-02	1.41E-01
	Aquatic marine	kg N eq.	3.01E-01	4.43E-03	4.23E-01	7.29E-01
	Aquatic terrestrial	mol N eq.	3.27E+00	4.81E-02	4.41E+00	7.73E+00
Photochemical oxidant creation potential (POCP)		kg NMVOC eq.	9.79E-01	2.18E-02	1.27E+00	2.27E+00
Abiotic depletion potential (ADP)*	Metals and minerals	kg Sb eq.	3.59E-03	1.37E-05	3.10E-04	3.92E-03
	Fossil resources	MJ, net calorific value	2.12E+03	7.05E+01	4.09E+03	6.29E+03
Water deprivation potential (WDP)*		m^3 world eq. deprived	1.85E+01	3.82E-01	1.50E+02	1.69E+02

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

Resource use indicators

PARAMETER	UNIT	A1	A2	A3	TOTAL
Use of renewable primary energy as energy carrier (PERE)	MJ, net calorific value	2.25E+02	1.10E+00	3.01E+02	5.27E+02
Use of renewable primary energy resources used as raw materials (PERM)	MJ, net calorific value	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Total use of renewable primary energy (PERT)	MJ, net calorific value	2.25E+02	1.10E+00	3.01E+02	5.27E+02
Use of non renewable primary energy as energy carrier (PENRE)	MJ, net calorific value	2.12E+03	7.05E+01	4.09E+03	6.29E+03
Use of non renewable primary energy resources used as raw materials (PENRM)	MJ, net calorific value	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Total use of non renewable primary energy resource (PENRT)	MJ, net calorific value	2.12E+03	7.05E+01	4.09E+03	6.29E+03
Use of secondary material (SM)	kg	1.13E+03	0.00E+00	0.00E+00	1.13E+03
Use of renewable secondary fuels (RSF)	MJ, net calorific value	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Use of non-renewable secondary fuels (NRSF)	MJ, net calorific value	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Net use of fresh water (FW)	m3	6.60E-01	1.14E-02	3.68E+00	4.35E+00
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water				

Waste indicators (optional)

PARAMETER	UNIT	A1	A2	A3	TOTAL
Hazardous waste disposed (HWD)	kg	1.28E-01	2.09E-03	1.58E-01	2.89E-01
Non-harzardous waste disposed (NHWD)	kg	5.68E+01	5.53E+00	1.13E+01	7.36E+01
Radioactive waste disposed (RWD)	kg	1.57E-03	1.66E-05	7.70E-03	9.29E-03

Output flow indicators (optional)

PARAMETER	UNIT	A1	A2	A3	TOTAL
Components for re-use (CRU)	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for recycling (MFR)	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for energy recovery (MER)	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported electrical energy (EEE)	MJ, net calorific value	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported thermal energy (EET)	MJ, net calorific value	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Disclaimer: “the environmental performance results are relative expressions and do not predict impacts on category endpoints, the exceeding of thresholds, safety margins or risks”.

ADDITIONAL ENVIRONMENTAL INFORMATION

Regulated Hazardous Substance

The base material of the steel plate is iron. No substances required to be reported as hazardous are associated with the production of this product.

Additional social and economic information

Not applicable

Information related to Sector EPD

This is an individual EPD.

VERSION HISTORY

This is the “original version of the EPD”, 2025-12-29

Abbreviations

CPC	Central product classification
c-PCR	Complementary product category rules
EAF	Electric Arc Furnace
EPD	Environmental product declaration
EU	European Union
GHG	Greenhouse gas
GHS	Globally harmonized system of classification and labelling of chemicals
GPI	General programme instructions
GWP	Global warming potential
ISO	International Organization for Standardization
LCA	Life cycle assessment
LCI	Life cycle inventory
LF	Ladle Furnace
ND	Not declared
PCR	Product category rules
PEF	Product environmental footprint
REACH	Restriction of chemicals
SI	The international system of units
SVHC	Substance of very high concern
VD	Vacuum Degassing

REFERENCES

1. ISO 14040:2006 Environmental management – Life cycle assessment-Principles and Framework

2. ISO 14044:2006 Environmental management – Life cycle assessment- Requirements and provides guidelines for life cycle assessment (LCA)

3. General Programme Instructions for the international EPD System, version 5.0.1

4. PCR 2015:03: Basic iron or steel products & special steels, except construction products, Version 3.0.0